ASYMMETRIC INFORMATION AND PRICE DISCOVERY IN THE FX MARKET: DOES TOKYO KNOW MORE ABOUT THE YEN?

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Abstract:
We identify a period in the foreign exchange market when there is a high concentration of informed yen/dollar traders active in Tokyo. We exploit the data during this period to test implications of market-microstructure theory. Comparing the period of informed trader clustering to a similar period without the informed, we find the following results:

1. **Exchange rate quotes adjust to full-information levels six times faster when the informed are active than when they are not.**
2. **Japanese quotes tend to lead the rest of the market when the informed are active. At other times, two-way causality is observed in quotes.**
3. **The contribution of Japanese quotes to yen/dollar price discovery relative to quotes of the rest-of-the-world is 5 to 12 percentage points higher when the informed are active compared to when they are absent.**

These results are consistent with a view of the foreign exchange market where private information is at times quite important, yet "normal" times are characterized as periods where public information, shared equally by all, results in a high contemporaneous correlation across quotes, regardless of origin.

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I. INTRODUCTION

If we could identify periods where a financial market experienced a concentration of informed trading activity and then compared this to other periods where such informed trading was less pronounced, what we would expect to find? Theory suggests we should observe important differences. Specifically, microstructure models of financial markets suggest that market dynamics will be affected by the presence or absence of informed traders (see O'Hara (1995) for a review). Empirical tests of such models are lacking due to the obvious problem of identifying an "informed" trader. We attempt to address this problem by focusing on a particular period of Asian yen/dollar trading when a compression of the time available for trading likely results in a high concentration of informed trading. By exploiting the information in yen/dollar quotes in this market interval, we have evidence supporting the hypothesis that there is, indeed, a high concentration of informed trading activity where the following empirical regularities are found: 1. Price adjusts faster to full-information levels during periods of concentrated informed trading. 2. The quotes of the informed traders tend to lead the quotes of the uninformed. 3. The contribution to price discovery of the quotes coming from the region where the informed are concentrated (Japan) is higher during times that are rich in informed trading than in other times.

The particular period of interest was identified in a recent paper by Ito, Lyons, and Melvin (1998), who studied a change in trading regime that occurred in the Tokyo foreign exchange market on December 22, 1994. On this day, Japan lifted a restriction against trading the yen over the lunch break from 12 to 1:30 (Tokyo time). Ito, et al. analyzed foreign exchange market activity during Asian business hours for the periods before and after the lunch-time trading and found evidence consistent with the presence of significant private information in the foreign exchange market. A particularly important piece of evidence in their study was that there is a U-

shape of volatility of yen/dollar exchange rate returns in the Tokyo morning when the lunchtime closing occurs that disappears once lunchtime trading begins. This is consistent with models of private information where the informed traders must exploit their informational advantage prior to the market closing so that a compressed trading time results in a faster revelation of the private information.

For our purposes, the late morning period in the pre-December 22, 1994 era (which will, henceforth, be identified as the BEFORE period) is viewed to be relatively rich in informed trading compared to the post-December 22, 1994 era (which will, henceforth, be identified as the AFTER period) when trading occurs through the midday lunchtime so no Tokyo market closing exists. In addition, we hypothesize that the Japanese traders have superior information on the yen/dollar exchange rate relative to non-Japanese traders.

A natural question at this point is to inquire about the source of this informational advantage. While one cannot offer any hard evidence on this question, there is wide recognition of some potential sources. One important source of private information in the foreign exchange market is the customer order flow. For instance, large Japanese banks will receive the foreign exchange business of the largest Japanese corporations and this proprietary information confers a temporary informational advantage on Japanese banks relative to non-Japanese banks in making a market in the yen/dollar. Another potential source of informational advantage is early knowledge of important government data releases or policy actions. Peiers (1997) presents evidence that Deutschebank acts as a price leader in the mark/dollar market around periods of Bundesbank intervention. The inference is that Deutschebank receives a significant portion of the intervention-related currency trades of Germany's central bank and this confers a temporary information advantage. This is but one possible example of how a large and important bank may have

temporary sources of private information. In similar fashion, it may be the case that some Japanese banks have superior ability to anticipate policy movements of the Japanese authorities.[1]

Some initial evidence supportive of the Japanese-informed-trader hypothesis is provided by the shape of volatility curves during the Asian business morning. Figure 1 depicts the pattern of the unconditional variance of yen/dollar quotes posted on the Reuters FXFX screen for each hour of the morning in the BEFORE period.[2] Quotes originating from Japan are seen to have a U-shape of volatility while quotes originating from other locations have a downward-sloping volatility pattern. As analyzed in Ito, Lyons, and Melvin, the U-shape of volatility is consistent with the presence of a concentration of informed traders prior to the lunch break in Tokyo trading. The informed must trade to exploit their informational advantage prior to the market closing. The lack of such a U-shape for non-Japanese volatility indicates no such information-based trading. Figure 2 displays the variance patterns over the Asian morning AFTER period. Note in this case, that the U-shape for Japan volatility has disappeared. With the market remaining open for lunch, trading (and information revelation) is spread out through time relative to the BEFORE period.

Figures 1 and 2 offer preliminary evidence consistent with a concentration of informed Japanese bank trading prior to the lunch break BEFORE. We will focus on this pre-lunch

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[1] An interesting and entertaining story told by one experienced trader is found in Niederhoffer's fable of "The Old Trader and the Yen" (in Niederhoffer,1997). He asserts the informational advantage of the Japanese banks when he describes them in the following passage: "At 7p.m. in New York, it is morning in Asia....The banks will wait for their customers to sell the dollar to hedge their export earnings, and then they will buy it....They drink sake each night with their friends from school and the ministries, and they learn what is going to be announced and where they will be buying and selling." Whether truth or fiction, if non-Japanese traders *believe* that Japanese banks have superior information, then the dynamics investigated in the following sections will be expected. Recently, a *Wall Street Journal* article discussed Japanese government investigation of corruption at the Bank of Japan and stated that "Traders have long accused central bank officials of leaking the *tankan* survey, a closely watched quarterly poll of business sentiment that can often suggest when the bank will change interest rates" (Sapsford, 1998, p. A12).

BEFORE period as a useful sample for testing implications of microstructure theories regarding market dynamics in periods rich in informed trading. Of course, the U-shape of volatility is not sufficient evidence of private-information-based trading. Such a U-shape could also be a result of inventory rebalancing at the open and close of trading as discussed in Brock and Kleidon (1992) so that the volatility is simply volume induced rather than associated with privately-informed trading. However, the following sections will present additional evidence of price-leadership and price discovery suggesting that Japanese banks are, indeed, better informed than the rest of the market. These results cannot be explained by simple inventory rebalancing stories.

[2] The data will be described in more detail in the next section. The variances were computed over 20 days for the BEFORE and AFTER periods using quotes sampled at one-minute intervals.

II. IMPLICATIONS OF INFORMED TRADER CONCENTRATION

Due to the lunch-time shutdown of trading in Tokyo prior to December 22, 1994, we expect that there was an exogenous concentration of informed trading prior to the 1.5 hour Tokyo market closure. What sort of market dynamics should result from this clustering of informed trading? Following the literature associated with papers such as Kyle (1985, 1989), Admati and Pfleiderer (1988), and Subrahmanyam (1991), in Section II.A we will briefly sketch the implications of a model for the foreign exchange market case under consideration. After presenting the model, we will provide an empirical analysis in Section II.B of one implication of the model—that a greater concentration of informed traders prior to lunch in the pre-December 22, 1994 period should be associated with a faster speed of adjustment of the yen/dollar exchange rate to full-information levels.[3]

II.A. Background Theory

The model has one risky currency traded by informed traders, liquidity traders, and a market maker. The exogenously-determined informed traders are assumed to be risk-averse and observe a private signal, δ, of the fundamental value of the currency. The liquidity traders trade for exogenously determined liquidity reasons, apart from fundamental value. The market maker is assumed to be risk neutral and can be thought of as a broker or intermediary who sets a price that yields zero expected profits given an observed net order flow. Trading occurs at time 0 and positions are liquidated at time 1. The terminal value of the currency is

[3] There are many different microstructure models that could be used to motivate alternative hypotheses. The one chosen is particularly nice for illustrating how the informational efficiency of price depends upon the number of informed traders—our first testable hypothesis. Alternatively, one could use an Admati and Pfleiderer type model to motivate the intradaily U-shape of volatility.

(1) $F = \bar{F} + \delta$.

$\bar{F}$ is public information at time 0 and δ is a random innovation to the fundamental value that becomes public at time 1. At time 0, informed traders observe a noisy signal, $\delta + u$, where $\text{var}(u) = \phi$ and $\text{var}(\delta)$ is normalized to equal 1 for simplicity. Both δ and u are mean zero, independently and normally distributed variables. There are k informed traders who observe identical signals and have common risk-aversion coefficient A and negative exponential utility. Liquidity trader demand is equal to z, a zero mean random variable distributed normally and independently of δ and u.

In the tradition of Kyle (1985), the market maker, using a linear pricing function, sets price P so that zero expected profits are earned conditional on the order flow:

(2) $P = \bar{F} + \lambda\omega$,

where ω is the net order flow and λ is the inverse of market depth (or liquidity). Subrahmanyam shows that in equilibrium, the following will be true:

(i) Each informed trader has a demand of $\beta(\delta + u)$ and

(3) $\beta = 1/\{(1+\phi)\lambda(k+1) + A[\phi + \lambda^2\sigma_z^2(1+\phi)]\}$,

where $\sigma_z^2 = \text{var}(z)$.

(ii) The equilibrium value of λ satisfies the quintic equation:

(4) $\lambda[k^2(1+\phi) + \sigma_Z^2\alpha^2] = k\alpha$,

where $\alpha = \beta^{-1} = A[\phi + \lambda^2(1+\phi)\sigma_z^2] + (k+1)\lambda(1+\phi)$.

Equation (3) indicates that, other things equal, informed traders demand increases when the precision of their private signal increases, the market is deeper, there are fewer informed traders,

and they are less risk averse. Equation (4) says that λ is a unimodal function of k, meaning that for k increasing from one to infinity, λ first increases, attains a peak and then decreases. Thus, the effect of changes in k on market depth or liquidity will depend on the magnitude of k. When k is small, the increase in k offsets the decrease in β and λ increases (market depth and liquidity falls). When k is big, the opposite is true. The interpretation of the case under consideration is that there is a high concentration of informed traders (large k) in the late morning prior to the lunch closure in Tokyo for the BEFORE period. While in the AFTER period, this informed trader clustering before lunch disappears. This suggests that λ may be decreasing (increasing) in k for the BEFORE (AFTER) period.

If the assumption of exogenous liquidity trading is relaxed and we consider a model of liquidity traders acting strategically, as in Admati and Pfleiderer, then the liquidity traders choose the timing of their trades in order to lower their costs of trading. Since trading costs are positively related to λ, we expect a higher concentration of liquidity trading in the BEFORE period (when λ is smaller) relative to the AFTER period (when λ is larger). An increase in liquidity trading BEFORE increases σ_z^2 and, via equation (4), reduces λ further. This concentration of trading activity for the BEFORE late morning can explain the U-shape of volatility that appears in the BEFORE morning.

One testable implication of informed trader concentration is the speed of adjustment of prices to full-information levels. Theoretically, the informational efficiency of prices (the degree to which price reveals private information) is defined as the posterior conditional variance of δ conditioned on the order flow and may be derived from equations (2) and (3). Let Q represent informational efficiency, then $Q = [\text{var}(\delta|\omega)]^{-1}$ or

$$(5)\ Q = 1 + \{1 / [\phi + (\sigma_z^2 / k^2\beta^2]\},$$

and we recall that β is an inverse function of k. In order to determine the effect of changes in k on Q, we must establish the effect of changes in k on $k\beta$. Multiplying equation (3) by k, it can be proved that $k\beta$ will increase with k. The intuition behind this result is the following: $k\beta$ measures the trading aggressiveness of informed traders. An increase in trading aggressiveness causes private information to be revealed in price faster and, hence, increases Q. So prices are more informative (and will converge more quickly to full information levels), the greater the number of informed traders in the market.

II.B. Speed of Adjustment to Full-Information Price

The previous section provided a sketch of a microstructure model where price discovery occurs more quickly during times of intense informed trader activity. Intuitively, the informed traders are revealing information to the rest of the market regarding unobservable fundamentals. In the case of the trading break during the Tokyo lunch time, we believe that there should be a clustering of informed traders prior to the market closure as traders seek to exploit any informational advantages prior to the closure. If this is the case, then we should observe a quicker pace of price discovery during this period of informed trader concentration.

To test this hypothesis, we examine the speed of adjustment of the yen/dollar exchange rate over the late morning (10:30am-12 noon) in Tokyo for both the BEFORE and AFTER periods. The data we use are based on quotes appearing on the Reuters FXFX screen during the period from November 1994 to January 1995. [4] Since the tick-by-tick data are well-known to

[4] The data were obtained from Olsen and Associates Research Institute for Applied Economics in Zurich. The sample periods for analysis are BEFORE: November 24 to December 21 and AFTER: January 4 to January 31.

have a low signal-to-noise ratio which can be improved by sampling at lower frequencies, we construct 1-minute returns.[5] The one minute returns were created by taking the change in the logs of the midpoints of the bid and ask quotes appearing prior to each minute.

A common method for estimating the speed of convergence of financial asset prices to "full-information values" is to estimate GARCH models and then compute the half-life of shocks -- the time it takes on average for the conditional variance of the series to revert halfway to its unconditional value. Specifically, we find (as do others before us) that the high frequency FX returns, r_t, are well represented by an MA(1)-GARCH (1,1) model of the following form:

$$(6)\ r_t = \alpha_0 + \varepsilon_t + \alpha_1 \varepsilon_{t-1}$$

$$(7)\ h_t = \gamma_0 + \gamma_1 h_{t-1} + \gamma_2 \varepsilon_{t-1}^2 + \gamma_3 dum * h_{t-1} + \gamma_4 dum * \varepsilon_{t-1}^2$$

where $\varepsilon \sim N(0, h_t)$ and the dummy variable, dum, equals 1 for the first observation each day and is zero otherwise. Estimates of the model are presented in Table 1. Note the significant moving-average term which captures the foreign exchange equivalent of the bid-ask bounce as traders shade quotes to one side of the market to maintain inventory balance. Note also the persistence in volatility seen in the high values of the lagged conditional variance parameters. Since the lagged residual squared and conditional variance reach back one period, the dummies for the first observation each day account for the break from the last observation yesterday. Taken as a whole, the estimation results reported in Table 1 look familiar in the context of previous GARCH models of high-frequency exchange rate data. Our interest in estimating these models is to estimate the speed of adjustment of prices.

The holiday period is omitted from the early AFTER sample as many traders take vacation and market activity is abnormally low.

[5] See Zhou (1996) for an analysis of optimal sampling frequency for tick-by-tick Reuters data.

To infer the speed of adjustment of price to its full-information value, we estimate the number of periods it takes for a shock to volatility to return half the way to the unconditional variance. The j-step ahead forecast of conditional variance implied by equation (7) is:

(8) $\underset{t}{E}[h_{t+j}] = \sigma^2 + (\gamma_1 + \gamma_2)^j (h_t - \sigma^2)$

where $\sigma^2 = \gamma_0 / (1 - \gamma_1 - \gamma_2)$ is the unconditional variance (unconditional expectation of h_t).[6] The half-life of volatility persistence is found by setting $(\gamma_1 + \gamma_2)^j = 1/2$ so that $j \ln(\gamma_1 + \gamma_2) = \ln(1/2)$ or the half-life of j is equal to $-\ln(2) / \ln(\gamma_1 + \gamma_2)$. The half-life estimates implied by the estimated models reported in Table 1 are 2.33 minutes in the BEFORE period and 14.09 minutes in the AFTER period.

The estimated half-lives are consistent with the hypothesis that there was a high concentration of informed trading in the Asian morning prior to lunch time when the lunch break occurred in Tokyo (the BEFORE period). This concentrated informed trader activity resulted in the yen/dollar exchange rate moving toward its full-information value at a faster pace than occurs in the absence of such informed trader presence. The late-morning clustering of the informed no longer occurred after the prohibition against lunch-time trading in Tokyo was lifted (the AFTER period) when the speed of adjustment of the exchange rate to its full-information level occurs at a slower pace.

[6] The expectation and the half-life analysis that follows is for all but the first observation of each day. In that case, the expectation and half-life will include the coefficients on the dummy variables.

III. JAPANESE AND NON-JAPANESE BANK DYNAMICS

As discussed in the introduction, it is commonly believed that Japanese banks have superior information regarding the yen. The source of this advantage is, presumably, the close links between banks and large non-bank corporate customers in Japan, who fill their currency needs through the Japanese banks. There may also be a closer relationship between Japanese governmental agencies and Japanese banks than foreign banks. Without being able to identify the source of informational advantage, if the rest of the market believes that Japanese banks are, at times, better informed on the yen/dollar exchange rate, we should expect to observe periods when the Japanese bank quotes *lead* the quotes of banks in other locations.

Section II provided some preliminary evidence that the late morning pre-lunch BEFORE period was a period of informed trading concentration, while the same time of day in the AFTER period contained no such clustering of the informed. Based on this finding, then if Japanese banks are believed to have superior information relative to non-Japanese banks, we hypothesize that the BEFORE pre-lunch period when informed trader clustering occurs in Japan, should exhibit Japanese bank quotes *causing* non-Japanese quotes, but this one-way causality should not appear in the AFTER pre-lunch period. This is, essentially, a test of whether the rest of the market believes that the informed yen traders are concentrated among the Japanese banks.

Since quotes from Japanese and non-Japanese banks arrive at different points in time and have a different number of observations in the sample we have a non-synchronous quoting problem. Any approach to this problem requires some potentially-controversial assumptions. Harris, et al. (1995) chose to use observations only when all markets have quotes, so their sample is defined by the least-frequent quote source. De Jong, Mahieu, and Schotman (1998) explicitly enter missing values for intervals when no new quote appears for a market and then develop an

estimator which explicitly accounts for these missing observations. We will utilize two approaches. First, we will filter the data for quote revisions that would be noticed in the noise of intradaily trading and use these big quote revisions to investigate causality. Second, we will employ the methodology of De Jong, Mahieu, and Schotman to utilize all quotes in inferring causality.

III.A. Large Quote Revisions and Causality

A strong test of causality would be provided by filtering the quotes to identify relatively large changes that should be noticed in the noise of intradaily quoting. Specifically, we filter the quotes according to whether they are entered by Japanese or non-Japanese sources and establish a minimum threshold for relevant quotes. We test for causality with the following regression:

$$(9)\ r_t^d = a + br_{t-1}^i + cr_{t-1}^d + e_t$$

where d and i index the origin of quotes as either Japan or non-Japan and r is the exchange rate return. So r_t^d is the return from consecutive quotes of region d at time t. When d is Japan and i is non-Japan, we test the null hypothesis that $b=0$, or non-Japanese returns do not cause Japanese returns. Similarly, when d is non-Japan and i is Japan, we test the null that Japanese returns do not cause non-Japanese returns. A lagged dependent variable is added to the equation to account for autocorrelation existing in the dependent variable series and guard against the independent variable return simply capturing the time-dependency that exists in high-frequency data. The model is estimated using a Newey-West adjustment for autocorrelation and heteroskedasticity.

The returns series are constructed from tick-by-tick data on the yen/$ for the same BEFORE and AFTER 20-day time windows identified above in Section II.B. Returns are defined

as the change in the log of the midpoint of the bid and ask quotes multiplied by 10,000 (so that the units are basis points). The data are constructed as follows:

(i) We begin by filtering the independent variable tick-by-tick returns so that only returns exceeding 2.5 basis points are kept. This filter is intended to reduce the noise in the series and sharpen the test so that independent variable observations are measuring sizable exchange rate changes that would be noticed in the noise of intradaily trading.

(ii) For each observation on the independent variable return, we construct the corresponding dependent variable return by taking the change in the log of the midpoint observations based on the dependent variable quotes following the independent variable returns.

The idea is to examine whether sizable changes in the quotes of one group have a significant effect on the quotes of the other group.[7]

Causality tests are performed separately for the BEFORE and AFTER samples. Results are given in Table 2. Note that there is two-way causality for every period except the late morning BEFORE. In this period, Japanese quotes are seen to significantly cause non-Japanese quotes in Table 2A. But in Table 2B, non-Japanese quotes do not significantly cause Japanese quotes for the same period. In every other period, both BEFORE and AFTER, two-way causality exists. We interpret these results as supportive of a market-wide perception of a concentration of Japanese informed traders prior to the lunch break in the BEFORE period. In this particular period, the actions of the Japanese traders attract the attention of the rest of the market and generate the price leadership behavior found in the data.

[7] While we later discuss implications of cointegration between Japanese and non-Japanese quotes when observations are sampled at 1-minute intervals, we point out that such cointegration issues are not relevant here since observations have been discarded through the filtering for large returns.

Note that the finding of two-way causality is consistent with a market driven by public information arrival. Public information shared by all would result in a general price adjustment across the market that may be spread over a short time. This brief period of quotes adjusting to reflect the new information would generate the two-way causality found in most periods. This is not to say that important private information is not also present during these periods, but rather the private information is not as dominant a factor in "normal" times. The evidence suggests that the Tokyo lunch time closing created an "abnormal" period of activity where private information revealed by the Tokyo traders dominated market dynamics in the period just prior to the lunch break in trading.

III.B. A Model of Non-Synchronous Quoting and Causality

Instead of filtering the data for large returns and then inferring causality, an alternative is to explicitly take into account the irregular spacing of the data and use all observations to investigate lead-lag relationships among Japanese and non-Japanese banks. There are two approaches to deal with irregularly spaced data: one which assumes that the most recent quote remains valid if the quote is not updated and another which explicitly puts a missing value at the intervals during which there is no new quote. We use the latter method with an estimator that takes these missing observations explicitly into account. This approach follows the contributions of Lo and MacKinlay (1990), De Jong and Nijman (1997), and De Jong, Mahieu, and Schotman (forthcoming). The method allows estimates of the cross-covariances between return series when the quotes are irregularly spaced in time.

We begin by assuming that the price setting process and the quote times are independent (timing is exogenous). An important feature of the method is that time is measured as calendar

(real) time and we assume that there exists a regular underlying unobserved discrete time process from which observed quotes are drawn. The ultimate goal is to estimate the cross-correlations between these unobserved returns.

The first step in the derivation is to write the difference between two observed quotes as a sum of the price changes of the underlying unobservable quote process:

$$(10)\ q_{t_{i+1}} - q_{t_i} = \sum_{t=t_i+1}^{t_{i+1}} \Delta q_t$$

where t_i denotes the clock time index of the ith observation. The unobserved quotes are sampled at the clock time frequency and the observed quotes are indexed by the quote time.

The expectation of the cross product of observed returns in the two markets can be written as :

$$(11)\ E(y_{ij}) = E[(q^J_{t_{i+1}} - q^J_{t_i})(q^N_{t_{j+1}} - q^N_{t_j})]$$

One can prove that the expectation of this cross product, conditional on the observed quote times, is a linear combination of the cross covariances γ_k of the underlying returns:

$$(12)\ E(y_{ij}) = \sum_{t=t_i+1}^{t_{i+1}} \sum_{k=t_j+1}^{t_{j+1}} \gamma_{t-k}$$

where k is the order of the cross covariances and

$$(13)\ \gamma_k = \mathrm{cov}(r^J_t, r^{NJ}_{t-k}) \qquad \text{where } r_t = q_t - q_{t-1}$$

Let $\chi_{ij}(k)$ be the number of times that γ_k appears in the right hand side of (12) and we can derive the following expression :

$$(14)\ \chi_{ij}(k) = \max[0, \min(t_{i+1}, t_{j+1} + k) - \max(t_i, t_j + k)]$$

An important property of (14) is that $\chi_{ij}(k)$ are functions of the quote times t only, and not of the observed quotes. Substituting (14) on the right hand side of (12) we obtain:

$$(15)\ E(y_{ij}\,|\,\chi_{ij}) = \sum_{k=-K}^{K} \chi_{ij}(k)\gamma_k$$

where we assume that all covariances of higher order than K are zero.

We can consider (15) as a regression with the unknown covariances γ_k as parameters and the coefficients $\chi_{ij}(k)$ as explanatory variables. In vector notation the regression is :

$$(16)\ y_{ij} = \chi_{ij}{}'\gamma + e_{ij}$$

The regression (16) can be estimated by OLS. In principle, the regression equation involves all possible cross products in the observed quotes changes . However, we can confine ourselves only to the adjacent quotes and we can ignore the observations that are more than K apart. De Jong and Nijman proved that the estimators of γ are consistent and the also provide a method to estimate the asymptotic standard errors of the estimators .

Estimates of the cross-correlations between Japanese and non-Japanese quotes are provided in Table 3A. These correlations are found as the estimated γ covariances from equation (16) divided by the standard deviation of the Japanese and non-Japanese returns over the same interval. Negative lags ($k>0$) represent the correlation between the current Japanese quote and non-Japanese quotes k periods ahead—so this tests whether Japanese quotes lead non-Japanese. Similarly, positive lags ($k<0$) test whether Japanese quotes lag non-Japanese. Wald tests for the joint significance of the leads and lags are reported in Table 3B. The evidence from the cross-correlations provided in Table 3A is somewhat difficult to interpret as the information in the individual significance of cross-correlations may be weak due to multicollinearity in the regression equation (16). The Wald tests of Table 3B send a clear signal: there is two-way causality in the

AFTER period, but in the BEFORE period there is one-way causality as Japanese quotes lead non-Japanese and the reverse causality is statistically insignificant.

Whether we use all the data, as just reported, or use only the observations involving large returns, we find the same result. In the BEFORE period presumed to be rich in informed trading, there is one-way causality where Japanese quotes lead non-Japanese. In the AFTER period, two-way causality exists regardless of which methodology is employed.

IV. PRICE DISCOVERY IN JAPAN AND ELSEWHERE

Having documented that non-Japanese quotes tended to lag behind Japanese quotes for the yen/dollar in the pre-lunch BEFORE period, we now want to conduct an analysis following Hasbrouck (1995) on the contribution of Japanese and non-Japanese quotes to yen/dollar price discovery. We begin by assuming that both Japanese (q_j) and non-Japanese (q_n) quotes contain a random-walk component (are I(1)) and the quote returns i.e., the differences of logs of the qs, are covariance stationary. These returns may be written in vector moving average form:

$$(17)\; r_t = \Psi(L)e_t$$

where $\Psi(L)$ is a polynomial in the lag operator and e_t is a zero-mean vector of serially uncorrelated innovations with covariance matrix Ω. While each quote series is nonstationary, the difference between the two series is stationary or the quotes are cointegrated of order 1. Then $\beta' q_t$, where $\beta' = [1 \; -1]$, is a stationary series, implying that $\beta'\Psi(1) = 0$, where $\Psi(1)$ is the sum of the moving-average coefficients. Similar to the Stock and Watson (1988) common trends representation of a cointegrated system, the rows of $\Psi(1)$ are identical. We denote ψ, as the common row vector in $\Psi(1)$, and it can be shown that the elements of ψ sum to unity.

As shown in Hasbrouck, the system may be written in error correction form as:

$$(18)\; r_t = \alpha(\beta' q_{t-1} - E\beta' q_t) + \Gamma_1 r_{t-1} + \Gamma_2 r_{t-2} + \ldots + \Gamma_{k-1} r_{t-k+1} + e_t$$

when there is a nonstationary autoregressive representation of order k for the quotes. The $E\beta' q_t$ term reflects systematic differences in the two quote series. In order to estimate this model, we follow Hasbrouck in estimating this term by calculating the mean value over the sample so that $(\beta' q_{t-1} - E\beta' q_t)$ is a "demeaned" quote vector. Once this vector is created, the model is then estimable by linear least squares.

If the error covariance matrix Ω is diagonal, then it is possible to directly calculate the contribution of the innovations in one market, say Japan, to the total variance. This may be thought of as the "information share" of the Japanese quotes. If only private information was relevant, we might expect such a structure. However, the presence of public information, shared equally by all, should lead to the innovations being contemporaneously correlated across markets. In this case, Ω is not diagonal. Hasbrouck recommends a procedure to bound the information shares of each region through a triangularization of Ω. First, let the innovations in the two regions' quotes be given by $e_t = Fz_t$ where F is the Cholesky factorization of Ω and z is a vector of random variables with $Ez_t=0$ and $Var(z_t)=I$ (an identity matrix). Then the market share of the innovation attributable to a particular z_j is found as:

$$(19)\ S_j = ([\psi F]_j)^2 / (\psi \Omega \psi')$$

where $[\psi F]_j$ is the jth element of the row matrix ψF. By permuting ψ and Ω, we get an upper (lower) bound on the information share of Japanese (non-Japanese) quotes by placing Japan first. When non-Japanese quotes are placed first, we get an upper (lower) bound on non-Japanese (Japanese) quotes. The factorization imposes a structure that maximizes the information share on the first quote and minimizes the information share of the last quote. As Hasbrouck states, using the Cholesky factorization in this manner is analogous to allocating the explained variance in a multiple regression by considering the incremental improvement in the R^2 that occurs when variables are added in sequence. The information share is a measure of who moves first in adjusting quotes. The magnitude of the difference between the upper and lower bound reflects the importance of contemporaneous correlation between Japanese and non-Japanese quotes.

With a diagonal Ω there is no such problem and unique values of the information shares may be estimated.

We estimate the information shares for the late-morning BEFORE and AFTER periods. The quotes are sampled at one-minute frequency, where the prior quote to each minute is used. No overnight returns are included. The specific functional form of equation (11) uses four lags of returns of each region. We experimented with alternative lags running from 2 to 12 and found that generally after 4 lags, the information shares are essentially unchanged. Table 4.A. contains the estimated bounds for the Japanese and non-Japanese information shares. We disaggregate all quotes into those originating in Japan and those originating elsewhere. Comparing the information share of Japanese quotes to those of non-Japanese quotes, we see in the BEFORE period that the Japanese information share is 96% of the non-Japanese (54%/56% or 44%/46%) at both the upper and lower bounds. In the AFTER period, the information share of Japanese quotes falls to 91% of non-Japanese at the upper bound and 84% at the lower bound. These results indicate that the BEFORE period of heavy Japanese informed-trader concentration has a larger share of information for Japanese quotes than in the AFTER period when the concentration of informed traders is absent.[8] This result is consistent with the hypothesis that the rest of the market is looking to Japan for information during the BEFORE period as the concentration of the informed Japanese traders tend to lead the market.

The fact that the information share of Japan in the BEFORE period is estimated to be slightly below that of the rest of the market at both the upper (54% to 56%) and lower (44% to

[8] As Hasbrouck states one can, in principle, compute an asymptotic standard error for the information share from the distribution of the estimated VAR parameters by linearization or Monte Carlo simulation. However, the information share is a nonlinear function of the VAR coefficients and the Ω matrix and the distribution of the latter is difficult to obtain. In this application, we will provide some evidence of robustness of the findings by estimating information shares for cross-sections of countries other than the Japan versus non-Japan sample.

46%) bounds is an artifact of the comparison of only one country's quotes, Japan, against all the rest of the quotes in the market. The methodology is sensitive to the frequency of quote updating and will be biased in favor of large information shares for a region with more frequent quoting. This is, of course, due to the fact that the more frequently quotes are posted, the smaller the number of one-minute observations with zero returns entering into the calculations. For this reason, we believe the relative information shares between Japanese and non-Japanese quotes are more useful to examine than the absolute estimates of the information shares of a single region.

To illustrate this quote frequency sensitivity, we estimated information shares of Japan versus Hong Kong in the BEFORE period, with results reported in Table 4.B. We now find that Japan's information share is 72% at the upper bound and 60% at the lower bound while Hong Kong's is 39% at the upper bound and 28% at the lower bound. The much less frequent quoting of Hong Kong compared to the entire non-Japanese region results in a much higher information share for Japan relative to Hong Kong than Japan relative to all non-Japanese quotes. Calculating the shares of information coming from Japan relative to Hong Kong, we find values of 181% at the upper bound and 213% at the lower bound in the BEFORE period. For the AFTER period, the shares are 174% and 208%. Thus, for the Japan/Hong Kong sample, we also find that the relative share of information coming from Japanese quotes falls in the AFTER period.

Finally, we compared quotes from Japan and Singapore in Table 4.C. Again, rather than focus on the absolute information shares contained in Table 4.C, we examine the relative information shares for inference. In the BEFORE period, Japan's information share is 109% of Singapore's at the upper bound and 110% at the lower bound. In the AFTER period, the relative information shares at the upper and lower bounds are 98% and 96%, respectively.

In each case of Table 4 we find that the relative share of information generated by Japanese quotes drops in the AFTER period relative to the BEFORE. Once the informed trader concentration ends, the importance of Japanese quotes relative to the rest of the market falls.

V. SUMMARY AND CONCLUSIONS

Microstructure models of financial markets have shown that we should expect markets to behave differently when there are many informed traders present than in times of informed trader absence. Empirical tests of the implications of these models have been hindered by the difficulty of identifying the informed traders. We have identified a period in the foreign exchange market where there should be a high concentration of informed trading. Following Ito, Lyons, and Melvin, we look to the late-morning period of yen/dollar trading in Tokyo prior to December 22, 1994 as a period where there is a concentration of informed Japanese traders relative to other times. This concentration of the informed arises from the closing of yen/dollar trading that occurred in Tokyo until December 22, 1994. After that date, Tokyo trading at lunch time was permitted. During the BEFORE period of lunch closing, Ito, Lyons, and Melvin provide evidence consistent with the existence of private information in the foreign exchange market where Japanese traders must trade prior to the lunch closing to exploit any informational advantage they possess. Once lunch trading occurs (the AFTER period), there is no longer any need for the informed Japanese traders to trade prior to noon and the evidence in Ito, Lyons, and Melvin indicates that the late-morning concentration of the informed no longer exists.

Based on this earlier work, we focus on the late-morning in Asian trading, hypothesizing that there should be different market dynamics in the BEFORE period compared to the AFTER period that would allow tests of the implications of market microstructure models. To this end, we investigate three separate hypotheses consistent with an informed trader concentration BEFORE that would disappear AFTER. A review of background theory following the spirit of the original paper by Kyle, provides a first hypothesis: the greater the number of informed traders, the more informative prices, so that prices will converge more quickly to full-information levels.

We test this hypothesis by estimating GARCH models of yen-dollar quote returns for the late-morning BEFORE and AFTER periods. Based on the model estimates, we infer the number of minutes it takes for a shock to volatility to return half way to the unconditional variance of quote returns. The half-life BEFORE is 2.33 minutes, while the half-life AFTER is 14.09 minutes. This finding that quotes adjust to full-information values six times faster in the BEFORE period than the AFTER period is strongly supportive of an important implication of microstructure theory.

If the market believes that Japanese traders are better informed regarding the yen, then any time there is a perceived concentration of the Japanese informed traders, we should expect to see quotes from Japan leading quotes of other regions. We tested this hypothesis by performing causality tests of Japanese versus non-Japanese quotes. In general, there is two-way causality. This is consistent with a market driven by public information arrival shared equally by all so that following the arrival of news, there is a brief period of quotes adjusting across the market to the new information. The general finding of two-way causality is also consistent with a market where private information is always important, but the informed behave strategically and reveal their information slowly over time so that in normal periods one cannot detect a dominant presence of the informed. The Tokyo lunch closing compressed the time available for the informed Japanese traders to exploit their informational advantage so that the period prior to the lunch break finds the informed traders forced to reveal their information more quickly. As a result, in this one period—the late-morning BEFORE—we find one-way causality where Japanese quotes lead non-Japanese quotes. In the period leading up to the Tokyo lunch time closing, the clustering of the informed Japanese traders seems to have created an environment where the rest of the market believed that the Japanese possessed superior information and were acting on that information so that Japanese quote leadership is observed.

A final examination of the implications of informed trader clustering was conducted by using Hasbrouck's methodology for estimating the contribution of different trading centers to price discovery. Specifically, we estimate the shares of information that may be attributed to Japanese quotes versus non-Japanese quotes. Given the contemporaneous correlation between Japanese and non-Japanese quotes, we are unable to estimate unique values of the information shares but, instead, estimate upper and lower bounds for each region. We find that in the late-morning BEFORE period the information share of Japanese quotes is 96% of the information share of the rest of the market. However, in the late-morning AFTER period, the information share of the Japanese quotes is between 91% and 84% of the rest of the market, depending on whether we evaluate at the upper or lower bound of estimates. We find qualitatively similar results comparing Japan with Hong Kong and Singapore. This drop in the information share of Japanese quotes in the AFTER period is consistent with the dissipation of the informed trader clustering existing in the BEFORE period. It appears that the Japanese traders' contributions to yen/dollar price discovery are at a peak when the market perceives these traders to have private information.

Taken as a whole, we have provided evidence on the implications of informed trader clustering in the foreign exchange market. While we cannot make any claims to generality of findings, it would be surprising if such implications did not extend to other financial markets. Returning to the question incorporated in the title: "Does Tokyo Know More About the Yen?", it appears the answer is a qualified "yes". At times, Japanese traders appear to have superior information—or, at least, the rest of the market believes that they do. However, the normal state of the market is one where Japan does not dominate in setting yen/dollar quotes. This does not mean that private information is unimportant in "normal" times, but rather that the informed are

able to slowly reveal their information. In this case, it may not be unreasonable to think of the market as largely being driven by public information.

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TABLE 1

GARCH Estimates of Yen/Dollar Returns for the Asian Late Morning and Implied Volatility Persistence *BEFORE* and *AFTER* Lunch Trading Begins

$$r_t = \alpha_0 + \varepsilon_t + \alpha_1 \varepsilon_{t-1}$$

$$h_t = \gamma_0 + \gamma_1 h_{t-1} + \gamma_2 \varepsilon^2_{t-1} + \gamma_3 dum * h_{t-1} + \gamma_4 dum * \varepsilon^2_{t-1}$$

Coefficient	*BEFORE*	*AFTER*
α_0	-0.013	-0.0007
	(-0.313)	(-0.028)
α_1	-0.293	-0.328
	(-8.144)	(-12.99)
γ_0	1.039	0.185
	(3.565)	(6.334)
γ_1	0.701	0.856
	(8.388)	(61.72)
γ_2	0.042	0.097
	(2.867)	(10.48)
γ_3	-0.498	0.386
	(-4.167)	(1.979)
γ_4	0.003	0.011
	(0.022)	(0.181)
Log likelihood	-3,746	-3,661
Skewness	10.66	0.37
Kurtosis	301.70	8.24
Q^2 (30)	11.00	16.99
Implied 1/2 life of shocks	2.33 minutes	14.09 minutes

Note: robust t-statistics in parentheses

TABLE 2

Causality Tests of Japanese and Non-Japanese Bank Quotes

$$r_t^d = a + br_{t-1}^i + cr_{t-1}^d + e_t$$

A. Non-Japanese Dependent Variable, Japanese Independent Variable

	a	b	c	q(12)
BEFORE 9-10:30	0.105 (0.86)	0.134 (4.06)	-0.219 (-2.83)	17.5
BEFORE 10:30-12	-0.379 (-2.30)	0.175 (4.36)	-0.529 (-5.48)	4.0
BEFORE 1:30-3	0.181 (1.22)	0.073 (2.77)	-0.135 (-1.36)	19.4
AFTER 9-10:30	-0.142 (-1.75)	0.123 (4.78)	-0.292 (-5.69)	9.3
AFTER 10:30-12	-0.002 (-0.02)	0.189 (5.48)	-0.422 (-6.99)	16.3
AFTER 1:30-3	0.027 (0.30)	0.180 (5.84)	-0.331 (-6.13)	16.1

B. Japanese Dependent Variable, Non-Japanese Independent Variable

	a	b	c	q(12)
BEFORE 9-10:30	0.028 (0.20)	0.121 (3.81)	-0.448 (-5.19)	10.3
BEFORE 10:30-12	0.119 (0.68)	0.048 (1.34)	-0.071 (-3.39)	9.8
BEFORE 1:30-3	-0.161 (-1.54)	0.107 (4.18)	-0.297 (-3.95)	7.4
AFTER 9-10:30	0.032 (0.39)	0.143 (5.35)	-0.348 (-7.30)	11.2
AFTER 10:30-12	-0.143 (-1.43)	0.128 (4.85)	-0.189 (-2.76)	4.9
AFTER 1:30-3	0.036 (0.40)	0.096 (3.51)	-0.266 (-4.57)	12.2

Note: t-statistics in parentheses

TABLE 3

Lead-Lag Correlations Between Japanese and Non-Japanese Quotes

A. Estimated Cross-Correlations

Lag	BEFORE	AFTER
-5	-0.287 (-1.41)	-0.033 (-0.25)
-4	0.385 (1.39)	-0.211 (-1.51)
-3	-0.253 (-1.10)	-0.038 (-0.26)
-2	-0.124 (-0.54)	0.103 (0.71)
-1	0.513 (2.44)	-0.322 (-2.45)
0	0.053 (0.22)	0.668 (4.54)
1	-0.262 (-0.89)	-0.003 (-0.02)
2	0.021 (0.08)	-0.039 (-.23)
3	0.066 (0.26)	-0.346 (-2.04)
4	-0.374 (-1.45)	-0.005 (-0.03)
5	0.254 (1.40)	0.044 (0.33)

Note: The table entries are estimated cross-correlations defined as $Cov(r^J_t, r^N_{t-k})$ divided by the standard deviations of r^J_t and r^N_t. Heteroskedasticity-consistent t-statistics are in parentheses.

B. Wald Tests for Lead-Lag Significance

	BEFORE	AFTER
Japanese quotes lead Non-Japanese	11.6	11.5
Non-Japanese quotes lead Japanese	7.1	14.6

Note: The table entries are F-statistics for the hypothesis that the lag or lead covariances are jointly equal to zero. This statistic has an asymptotic distribution of $\chi^2(5)$.

Table 4

Information Shares of Japanese and non-Japanese Quotes

A. Japan and non-Japan Quotes

		Bounds for Information Shares			
Quote Origin	BEFORE:	Upper	Lower	AFTER:	Upper	Lower
Japan		54%	44%		62%	32%
non-Japan		56%	46%		68%	38%

B. Japan and Honk Kong Quotes

		Bounds for Information Shares			
Quote Origin	BEFORE:	Upper	Lower	AFTER:	Upper	Lower
Japan		72%	60%		72%	59%
Hong Kong		39%	28%		41%	28%

C. Japan and Singapore Quotes

		Bounds for Information Shares			
Quote Origin	BEFORE:	Upper	Lower	AFTER:	Upper	Lower
Japan		55%	50%		65%	34%
Singapore		50%	45%		66%	36%

Figure 1: Variance of Yen/Dollar Quotes in Asian Morning -- BEFORE

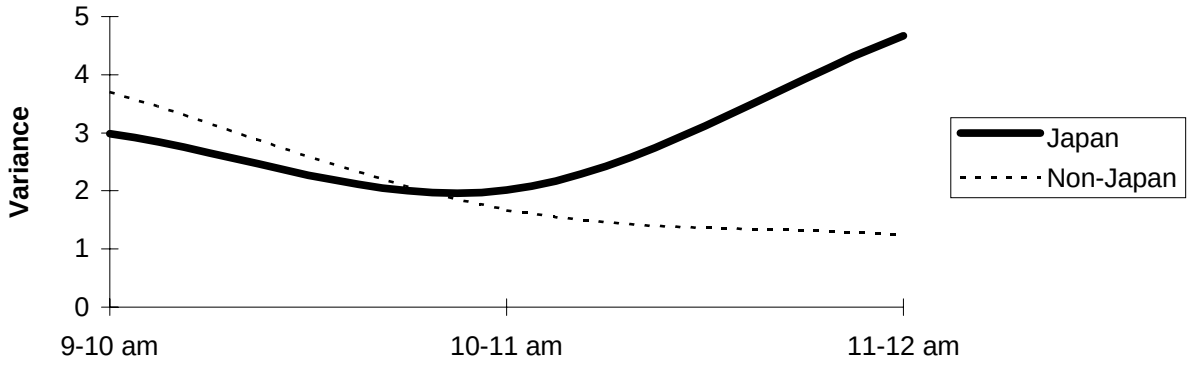


Figure 2: Variance of Yen/Dollar Quotes in Asian Morning -- AFTER

